

Signal Processing

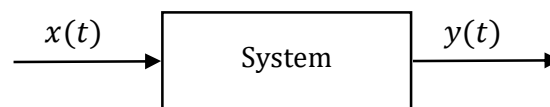
Exercise 3

Hints - Linear Time-Invariant Systems

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System

A system $\mathfrak{S}(x(t))$ assigns a function/an output signal $y(t)$ to a function/an input signal $x(t)$.



Linearity

Knowing the system responses $y_1(t)$ and $y_2(t)$ and its corresponding linearly independent input signals $x_1(t)$ and $x_2(t)$ allows to determine whether a system is linear or non-linear.

A system $\mathfrak{S}(x(t))$ is considered to be linear if both the principle of **superposition** and **homogeneity** hold.

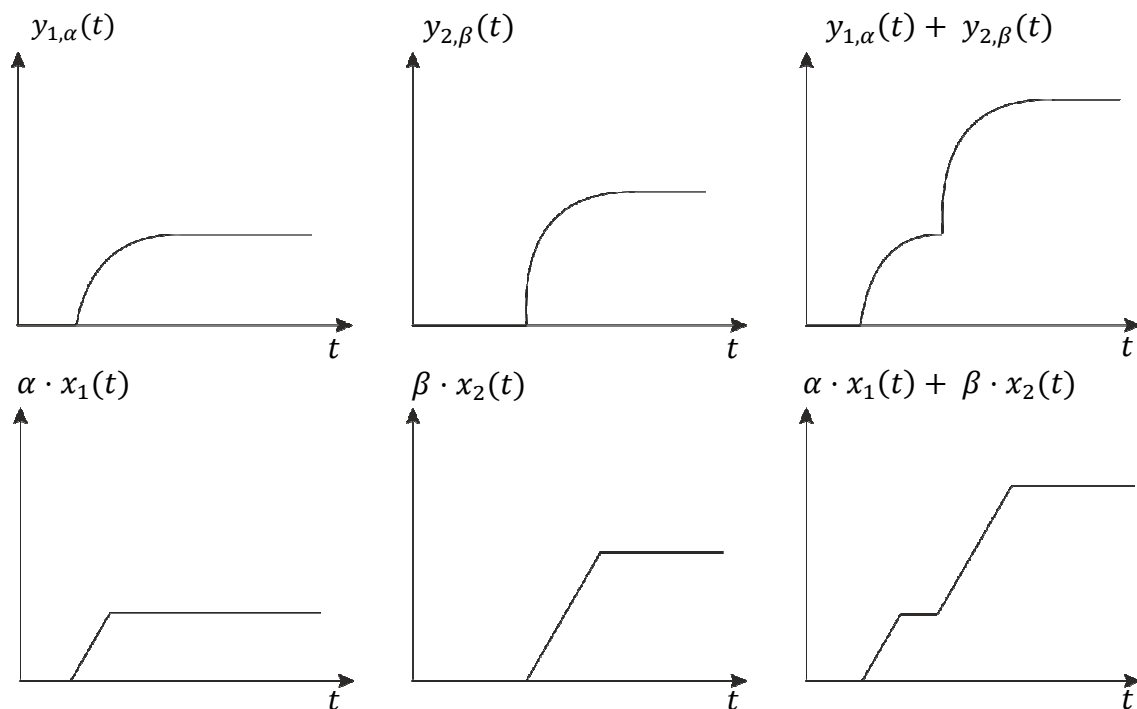
Superposition:	$x_1(t) + x_2(t) \rightarrow y_1(t) + y_2(t)$
Homogeneity:	$\alpha \cdot x(t) \rightarrow \alpha \cdot y(t) \quad (\alpha: \text{constant})$

This implies that a system $\mathfrak{S}(x(t))$ is linear if an input signal $x_i(t)$, which consists of a linear combination of two independent signals $\alpha \cdot x_1(t)$ and $\beta \cdot x_2(t)$

$$x_i(t) = \alpha \cdot x_1(t) + \beta \cdot x_2(t), \quad (1)$$

results in a system response $y_r(t)$ that comprises a linear combination of the system responses $\alpha \cdot y_1(t)$ and $\beta \cdot y_2(t)$ of each input signal $\alpha \cdot x_1(t)$ and $\beta \cdot x_2(t)$

$$y_r(t) = y_{1,\alpha}(t) + y_{2,\beta}(t). \quad (2)$$



Step-By-Step Guide to Prove Linearity:

1. Insert a linear combination $x_i = \alpha \cdot x_1(t) + \beta \cdot x_2(t) \rightarrow \mathfrak{S}(\alpha \cdot x_1(t) + \beta \cdot x_2(t))$.
2. Determine what the output signal $y_r(t)$ is supposed to look like if the system were linear $y_r(t) = y_{1,\alpha}(t) + y_{2,\beta}(t)$.
3. The system is only linear if the results of step (1) and step (2) are identical.

Time-Invariance

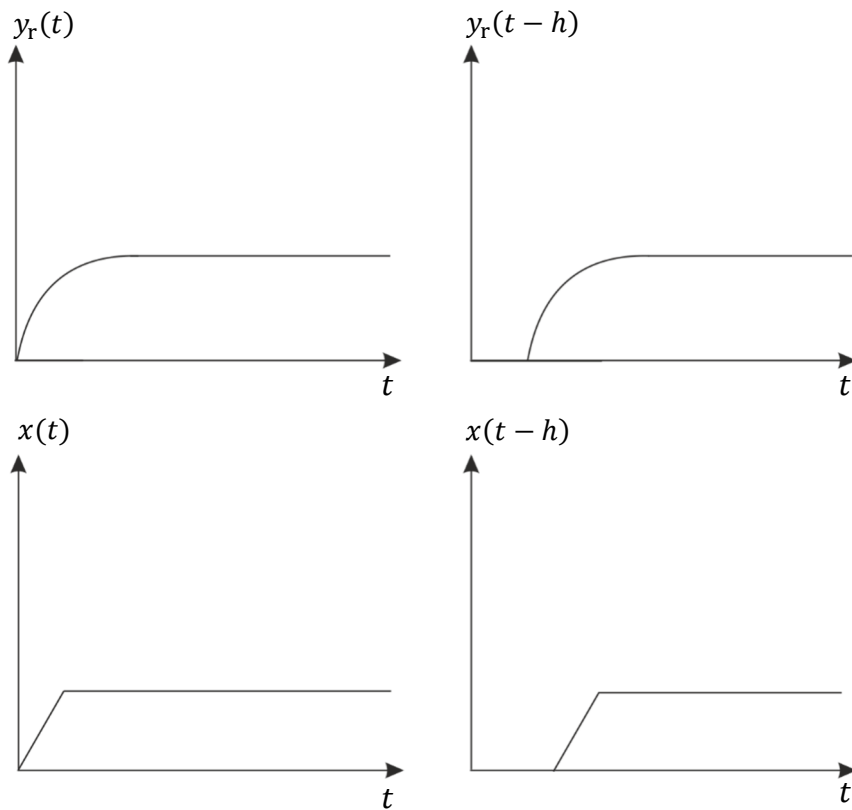
A system $\mathfrak{S}(x(t))$ is time-invariant if the system response $y_r(t)$ is delayed to the same extent as the input signal $x(t)$. This implies

$$x(t - h) \rightarrow y_r(t - h), \quad (3)$$

where $h \geq 0$.

Step-By-Step Guide to Prove Time-Invariance:

1. Delay input signal and determine the system response.
2. Determine a systems response if the time delay is only applied to $y_r(t)$.
3. The system is time-invariant if the results of both step (1) and (2) are identical.



Linear Time-Invariant System

We consider a system $\mathfrak{S}(x(t))$ to be an LTI-system (linear time-invariant system) if the system is both linear and time-invariant.

Causality/Causation

A system $\mathfrak{S}(x(t))$ is considered to be causal if it only depends on current or past values of $x(t)$. This implies $t \leq t_0$ for which t_0 represents the present time.

Nothing happens without a cause!